A provenance-tracked candidate dataset for Portuguese 60 kV topology reconstruction

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Scientific Data draft; public-source validation route

Abstract

Public distribution-network records rarely provide explicit bus–branch topology. We present PT60-Candidate, a provenance-tracked candidate dataset for the Portuguese E-REDES 60 kV layer and its fail-closed construction pipeline. From 5,334 line features and 586 facility records, the selected configuration yields 1,342 classified circuit candidates: 358 retained inter-facility branches and 984 downgraded or rejected records with reasons. Version 1.0.1 includes a 484-node multigraph, 216-configuration sensitivity sweep, source manifests, corrected schemas and data dictionary, checksums, and branch-level public-source evidence. Internal validation tests geometry, identifiers, graph–table parity, schema coverage, missingness, and archive integrity. OpenStreetMap/OpenInfraMap triangulation provides inspectable name or corridor evidence for every retained branch; endpoint-name permutation and displaced-geometry controls reduce the corresponding evidence rates by 95.1% and 99.7%. These results test matcher selectivity and public-data concordance, not physical topology accuracy. The dataset is not operator validated, precision and recall are unavailable, and electrical parameters remain incomplete. It supports topology-reconstruction, provenance, geospatial-integration, and interface research, not operational grid studies. Versioned data and code are provided separately.

Keywords: power-grid dataset; data provenance; topology reconstruction; distribution networks; technical validation; open data

1 Background & Summary

Public distribution-network portals can support geospatial and data-integration studies, but the released records rarely correspond directly to an auditable bus–branch dataset. Connectivity must usually be inferred from facility locations and line geometries, while electrical parameters, switching states, and operating conditions remain incomplete or unavailable. In this setting, a reusable dataset should not only publish retained candidate connections; it should also preserve how those candidates were constructed, what evidence supports them, and which fields remain unknown or diagnostic.

PT60-Candidate was assembled to provide such a resource for a Portuguese E-REDES 60 kV layer. The release is explicitly a *candidate* dataset. It does not represent an operator-confirmed connectivity model, a national Portuguese grid model, or a validated electrical operating case. Instead, it provides a provenance-tracked reconstruction product in which retained branches, downgraded circuits, release manifests, validation outputs, and downstream interface contracts remain inspectable as part of the data record.

The release contains three closely related resource types. First, it provides a core topology layer composed of retained inter-facility branches, an all-facility multigraph, and a complete circuit-classification ledger. Second, it provides validation and provenance records, including a reconstruction-sensitivity sweep, a source manifest, public-source audits, negative controls, and branch-level triangulation outputs. Third, it provides a schema package, field-level data dictionary, join documentation, manifest, checksums, and responsible-use metadata so that the core records can be interpreted without the development repository.

This framing differs from a conventional engineering paper. The central contribution is the release of a documented candidate dataset and its validation boundaries, not a claim of algorithmic novelty or a demonstration of predictive superiority. Prior open-grid resources have shown the value of transparent power-system model construction from public records [Medjroubi et al., 2017, Rivera et al., 2019, Hörsch et al., 2018, Xiong et al., 2025, Wiese et al., 2019]. PT60-Candidate extends that tradition to a Portuguese 60 kV distribution-layer reconstruction problem while making rejected circuits, semantic status fields, and downstream release contracts explicit.

2 Methods

2.1 Source datasets

The source snapshot combines AT line geometries, AT substations, switching facilities, MT-substation context, network-characteristics tables, substation-load records, reception-capacity context, and load-diagram indices from the E-REDES Open Data Portal [E-REDES, 2026]. The topology-critical inputs are `rede-at-teste` (5,334 records; checked 2026-07-13 14:15:57 UTC), `se-at_2025` (410 records; checked 2026-07-13 14:16:08 UTC), and `pc-at_2025` (76 records; checked 2026-07-13 14:15:31 UTC); their catalog, portal, CSV-export, and GeoJSON-export URLs are preserved in the source manifest. For the documented snapshot, the topology stage loads 586 technically relevant facilities, comprising 409 AT substations and 177 additional facilities. The same snapshot contains 5,334 valid AT line features, of which 5,323 are labeled 60 kV and 11 are labeled 130 kV in the current local build.

The release uses the portal-level license statement reported by E-REDES for distributable data artifacts. The dataset package records publisher attribution, portal URL, source dataset identifiers, access dates, and transformation notices separately from the repository's MIT code license. Where topology-critical API catalog responses did not repeat an explicit license field, the release policy follows the recorded portal-level statement rather than inferring a different license from the omission. The topology-critical catalog records were checked on 13 July 2026; per-source timestamps are preserved in `provenance/reproduction_source_manifest.json`.

2.2 Data acquisition

Source files are downloaded and validated through a reproducible workflow that records dataset roles, retrieval checks, and release metadata in a source manifest. The source manifest currently covers 13 datasets and is intended to accompany any public release. Geometry parsing and machine-readability checks occur before reconstruction so that parse failures become explicit blocking states rather than silent exclusions.

The acquisition scope is limited to the E-REDES AT layer represented in the documented snapshot. It does not include a verified REN transmission boundary, complete national generation, measured operating conditions, or authoritative switching-state data. These omissions are treated as scope boundaries of the dataset rather than as gaps to be silently imputed.

2.3 Data preprocessing

Preprocessing harmonizes source geometries, facility identifiers, and facility classes before topology inference. The workflow validates geometry type and validity, repairs parseable geometry issues where permitted by the loader, performs coordinate-reference conversions needed for endpoint and corridor calculations, and derives geometry-based quantities such as length. These transformations are recorded as derived rather than observed values.

Facilities are standardized into node candidates with associated codes, names, types, source roles, and footprint-ready geometries. Source line features are standardized into endpoint-bearing line records. Ambiguous matches and unresolved endpoints are retained as explicit states for later classification. This preprocessing stage therefore prepares line and facility inputs for reconstruction without asserting that each input line already corresponds to a valid inter-facility branch.

Table 1: Semantic statuses used in PT60-Candidate and its derivative interfaces.

Status	Interpretation
Observed	Copied from an E-REDES record, subject to source and join quality.
Derived	Deterministic transformation, such as geometry length.
Inferred	Connectivity produced by matching, clustering, merging, or classification.
Source-backed	Supported by an applicable operator, standard, or catalog source.
Scenario-assumed	Explicit choice used only in a named diagnostic scenario.
Benchmark-only	Non-Portuguese reference value used only for software and interface testing.
Proxy	Surrogate identity, injection, capacity, dispatch, or cost semantics.
Missing	Intentionally unfilled; dependent readiness gates must block use.

2.4 Provenance model

The release uses a field-level semantic status model so that downstream users can separate copied observations from deterministic derivations, inferred topology, and diagnostic assumptions. For an object o at stage k, the workflow records

$$T_k(o) = (v, s, q, a, e), \tag{1}$$

where v is the value, s its semantic status, q a quality or confidence indicator, a a source or assumption identifier, and e an exclusion or blocking reason.

Table 1 summarizes the evidence distinctions carried into the released records and metadata so that consumers can apply explicit admissibility rules. The complete field-level dictionary is provided as `schema/data_dictionary.csv`; `schema/file_schema_summary.csv`, `schema/join_relationships.csv`, and `manifest.json` provide file-level structure, joins, versions, roles, and checksums.

2.5 Reconstruction pipeline

Facility records are converted to circular footprints with radii of 50, 100, 250, and 500 m. Three node sets are evaluated: set A contains 409 AT substations, set B contains 484 AT substations and switching facilities, and set C contains all 586 technically relevant facilities. Each source line contributes two endpoints. When an endpoint intersects multiple footprints, the workflow resolves membership by centroid distance and records ambiguity where needed rather than forcing an unqualified assignment.

Endpoints are then clustered at 0.5, 1, 5, 10, 25, and 50 m. Union–find merging groups features into circuit candidates under geometry-only, voltage-aware, or voltage-plus-status-aware compatibility rules. Explicit conflicts block merging in the corresponding compatibility mode. Endpoint-cluster degree is then used to distinguish ordinary terminal pairs from loop, self-loop, isolated, and tap or multi-terminal patterns.

Figure 1 shows how retained and downgraded records pass through common validation and release-contract stages.

2.6 Candidate selection

A merged group is retained as an inter-facility candidate branch only when it has exactly two terminals, both terminals match facilities without disqualifying ambiguity, and the two facilities are distinct. Outcomes failing these conditions remain available in the circuit-classification ledger as loop, self-loop, single-facility, isolated, tap or multi-terminal, ambiguous, or otherwise downgraded records.

This selection policy is intentionally conservative. It favors explicit preservation of non-retained outcomes over optimistic retention. Mixed voltage or status conflicts, suspicious spatial matches, and unresolved facility mappings are retained as audit information instead of being absorbed into a simplified final graph.

2.7 Parameter sweep

The reconstruction workflow evaluates 216 settings generated by combining three facility sets, four footprint buffers, six endpoint snap thresholds, and three merge modes. The selection rule rewards retained inter-facility yield and interpretable component structure while penalizing ambiguity, explicit conflicts, suspicious

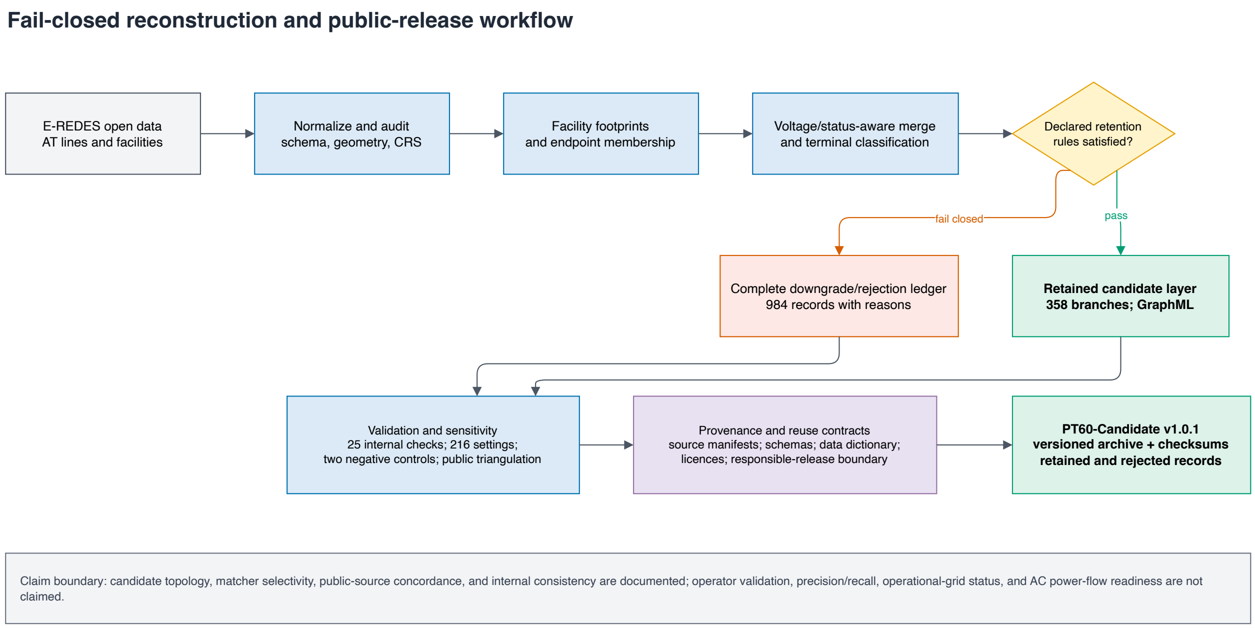

Figure 1: Fail-closed reconstruction and release workflow for PT60-Candidate. Blue boxes denote transformations or validation, the yellow diamond is the declared retention gate, green boxes are deposited release layers, the red box preserves downgraded/rejected records, and the purple box records provenance and reuse contracts. The diagram summarizes workflow structure and claim boundaries rather than measured performance.

matches, and the broadest thresholds. The selected release setting uses node set B, a 100 m facility buffer, a 0.5 m endpoint snap threshold, and voltage-plus-status-aware merging.

The score is a reproducible engineering rule used to choose one documented release point from the 216-setting sweep. It is not a calibrated probability of correctness and is not presented as an external optimum.

Large-language-model assistance was used for code review, structural editing, and release-checklist support. The authors inspected the generated changes and remain responsible for the methods, numerical results, citations, release files, and claims. This disclosure must be confirmed against the authors' actual use before submission.

2.8 Dataset generation

The selected setting converts 5,334 source line features into 1,342 circuit candidates. Of these, 358 satisfy the declared inter-facility rules and are released as retained candidate branches. The remaining 984 circuit candidates are preserved as downgraded records rather than discarded. The selected all-facility GraphML preserves isolated facilities as nodes so that graph outputs remain consistent with the declared isolate policy.

The same workflow packages the reconstruction-sensitivity table, public-source triangulation outputs, negative controls, an independence-risk audit, internal-validation results, source manifests, schemas, and checksum-linked inventory. Electrical diagnostics and Stage 1–5 learning interfaces are excluded from the main v1.0.1 public archive because they depend on incomplete, proxy, benchmark-only, or scenario-derived information.

3 Data Records

3.1 Repository overview

PT60-Candidate version 1.0.1 has been prepared for a reserved figshare record. Its DOI is `https://doi.org/10.6084/m9.figshare.32984021`. The archive `PT60-Candidate-v1.0.1.tar.gz` has SHA-256 `a4dd`

Table 2: Directory structure of the PT60-CANDIDATE v1.0.1 archive. Counts refer to files, not data rows.

Path	Files	Contents
`core_topology/`	12	Retained branches, complete circuit ledger, GraphML, sensitivity sweep, endpoint/facility summaries, voltage inventory, and parameter-readiness summary.
`provenance/`	5	Source manifests, API validation, licence decisions, and responsible-release boundary.
`validation/`	13	Internal checks, missingness, two negative controls, public-source concordance, and sanitized independence-risk records.
`schema/`	22	Field dictionary, file schemas, joins, CRS/geometry documentation, and 15 machine-readable schemas.
`inventory/`	1	Frozen headline counts and release-scope statements.
Archive root	10	Readme, citation, licences, attribution, changelog, manifest, checksums, exclusions, and validation summary.

`2d115caae0cd1c873fb7db3341e0aaed0ce62f7f0072ae92642898efb69a`. The v1.0.1 file must replace the earlier uploaded draft before the item is published.

The archive contains 63 documented files. It provides the retained candidate topology, the complete downgrade/rejection ledger, the GraphML multigraph, the 216-setting sensitivity record, provenance manifests, validation outputs, schemas, licensing documents, and checksum-linked inventory. Raw E-REDES downloads, raw OSM history/cache blobs, manuscript work files, optional Stage 1–5 interfaces, and non-operational electrical diagnostics are not part of the main public archive. The reasons and reproducibility consequences of these exclusions are recorded in `excluded_artifacts.json`.

3.2 File organization

The top-level `README.md`, `CITATION.cff`, `DATA_LICENSE.md`, `ATTRIBUTION.md`, `CHANGELOG.md`, and `LICENSE-CODE-MIT` describe citation, licensing, attribution, version history, and scope. The archive-control files `manifest.json`, `checksums.sha256`, `archive_validation_summary.json`, and `excluded_artifacts.json` identify every released file and verify package integrity. Table 2 summarizes the directory-level organization.

3.3 Released artifacts

Table 3 identifies the principal machine-readable records and their roles.

Retained candidate branches CSV. The retained branch table (`core_topology/at_interfacility_candidate_branches.csv`) is the principal branch-level record. Its 358 rows contain generated branch and circuit identifiers, endpoint facility identifiers and labels, voltage and status fields, geometry-derived length, segment count, source-line membership, confidence metadata, and explicit electrical placeholders. Fields ending in `_m` and `_km` use metres and kilometres, respectively. Missing electrical fields are not backfilled with model defaults.

Circuit-classification ledger CSV. The circuit-classification ledger (`core_topology/at_circuit_classification.csv`) contains all 1,342 merged circuit candidates, including the 358 retained records and 984 downgraded or rejected outcomes. Rows include construction-setting identifiers, terminal structure, facility membership, ambiguity and conflict indicators, source-line membership, disposition, and downgrade reason. The ledger exposes what was not retained and why.

Reconstruction-sensitivity CSV. The sensitivity table (`core_topology/at_paper_logic_parameter_sweep.csv`) stores all 216 combinations of facility set, facility buffer, endpoint snap threshold, and merge mode together with the corresponding reconstruction summaries. The selected setting is identified in `core_topology/at_paper_logic_summary.json`.

Table 3: Principal machine-readable artifacts in the deposited archive. Paths are relative to the archive root.

Artifact	Rows/items	Purpose
Retained candidate branches CSV	358	Inter-facility candidate branches with endpoint facilities, source identifiers, geometry-derived length, status, and confidence metadata.
Circuit-classification ledger CSV	1,342	Full retained and downgraded circuit population with terminal structure, conflict flags, and disposition fields.
Selected GraphML multigraph	484 nodes, 358 edges	All-facility candidate graph including isolates under the declared release policy.
Reconstruction-sensitivity CSV	216	Parameter sweep across facility sets, buffers, snap thresholds, and merge modes.
Public-source concordance CSV/JSON	358 branch records	Public-evidence categories, matching variables, source roles, and supporting summaries.
Negative-control CSV/JSON	2 controls	Endpoint-name corruption and spatial-displacement matcher-selectivity results.
Source manifests CSV/JSON	13 sources	Source dataset identifiers, roles, access checks, and attribution metadata.
Data dictionary CSV	1,090 records	Field, JSON-pointer, and GraphML-attribute definitions across 54 machine-readable paths.

GraphML multigraph. The GraphML file (`core_topology/at_paper_logic_graph.graphml`) represents the selected candidate multigraph. Nodes correspond to facilities in the selected release node set, including declared isolates; edges correspond to retained candidate branches. The `branch_id` edge identifiers and facility identifiers link the GraphML to the retained branch table and facility summaries. Parallel edges are preserved.

Validation and public-evidence records. The `validation/` directory contains the 25 internal checks and missingness report; endpoint-name and spatial-alignment negative-control records; branch-level OSM/OpenInfraMap concordance; a source-role audit; and a sanitized independence-risk audit. The `branch_id` field links branch-level validation records to the retained branch table. Evidence categories and independence-risk fields are audit variables, not confirmed topology labels.

Metadata and manifests. The parallel `provenance/reproduction_source_manifest.csv` and `.json` files record dataset identifiers, roles, access checks, and attribution metadata for 13 sources. The release-level `manifest.json` contains relative path, purpose, semantic role, media type, bytes, SHA-256, row/item count, dataset and schema versions, generator commit, upstream source identifiers, licence class, and public-access class for each file. `checksums.sha256` provides an independently checkable path–digest list.

3.4 Schema, coordinate systems, units, and missing values

The schema package documents 54 public machine-readable paths using 1,090 field, JSON-pointer, and GraphML-attribute records. `schema/data_dictionary.csv` defines type, nullability, unit, observed missing representation, allowed values, semantic status, derivation, and key role. Version 1.0.1 corrects the earlier substring-based inference of coordinate, geometry, boolean, and count fields and adds fail-closed semantic checks plus regression tests for those failure modes. `schema/join_relationships.csv` defines six recommended joins among retained branches, the ledger, validation records, source manifests, and archive checksums. Fifteen JSON schema files cover the principal core, validation, provenance, manifest, and GraphML records.

Released geometries use GeoJSON-style WGS 84 coordinates (EPSG:4326) in longitude–latitude order. Endpoint clustering, facility distances, corridor distances, and length diagnostics use a pipeline-defined local equirectangular metric workspace centred at longitude -8.532604 and latitude 39.567953; this workspace has no EPSG identifier. Voltage fields preserve file-specific source encodings. Common missing representations

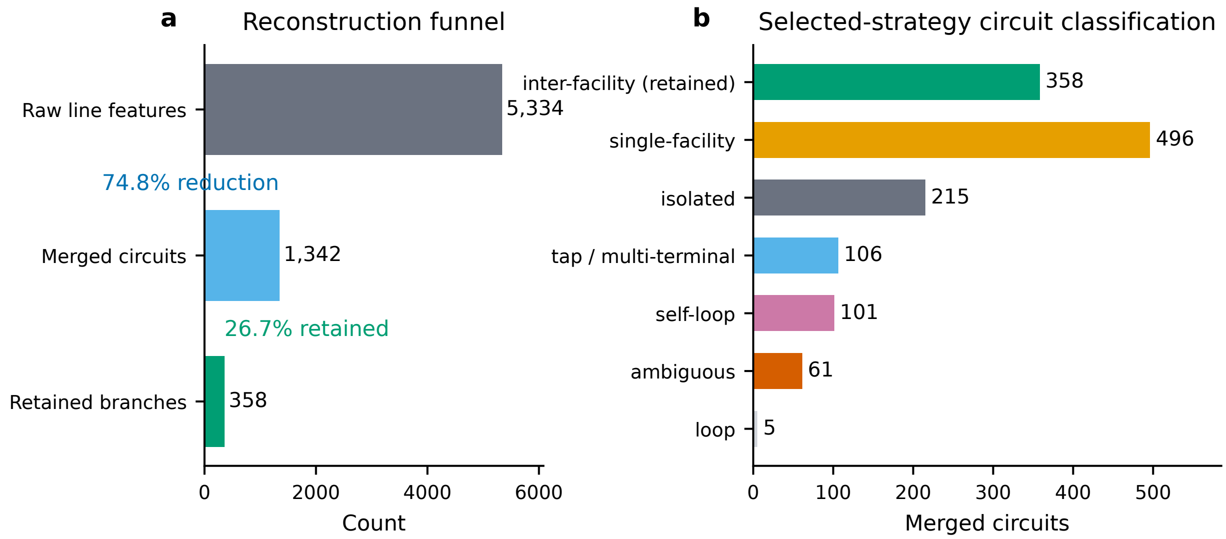

Figure 2: Record disposition for the selected reconstruction setting. The workflow processes 5,334 source line features into 1,342 circuit candidates, retains 358 inter-facility candidate branches, and preserves downgraded classes in the ledger. Retention denotes rule satisfaction under the documented workflow, not operator confirmation.

are empty strings, JSON `null`, `NaN`, absent optional keys, and the explicit token `MISSING_NOT_ESTIMATED`. The complete conventions are recorded in `schema/crs_and_geometry.md` and its JSON equivalent.

4 Data Overview

For the selected setting, the release accounts for all 1,342 reconstructed circuit candidates: 358 are retained as inter-facility candidate branches and 984 remain in the ledger with downgrade or rejection dispositions. Figure 2 reports the complete disposition, and Fig. 3 reports aggregate graph structure. This overview describes the contents of the deposited files; it does not estimate the correctness or completeness of the physical Portuguese network.

5 Technical Validation

5.1 Internal validation

The workflow applies fail-closed checks throughout acquisition, reconstruction, validation, and packaging. These checks include geometry-type and geometry-validity screening, coordinate-reference handling, endpoint completeness, facility-membership ambiguity, endpoint-cluster composition, mixed voltage or status conflicts, terminal-count classification, branch-key uniqueness, graph/table parity, schema coverage, manifest coverage, and checksum reconciliation.

Archive-level validation additionally checks that all 63 released files are documented, all manifest and checksum paths resolve, all 54 machine-readable paths have dictionary coverage, and the frozen headline counts agree with the core records. These results support internal usability of the deposited records, but they do not constitute external topology or electrical validation.

5.2 Structural validation

Structural validation focuses on whether the reconstructed records satisfy the declared topology-building rules. Under the selected setting, the workflow converts 5,334 line features into 1,342 merged circuit candi-

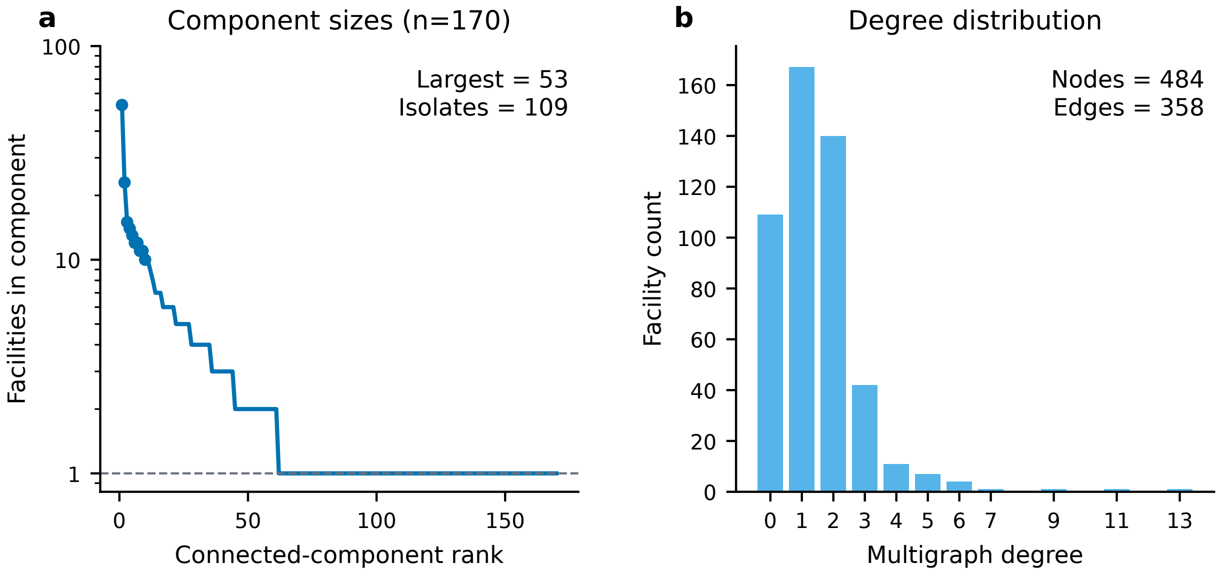

Figure 3: Aggregate structure of the selected candidate multigraph. Component sizes are shown on a logarithmic vertical scale and the degree histogram retains parallel edges. The figure describes the deposited graph artifact rather than the physical Portuguese network.

dates, of which 358 are retained and 984 are downgraded. The downgraded outcomes comprise 496 single-facility records, 215 isolated records, 106 tap or multi-terminal records, 101 self-loops, 61 ambiguous records, and five loop records for the current snapshot. Preservation of these downgraded classes is itself a validation feature because it prevents silent loss of evidence about reconstruction uncertainty.

5.3 Schema validation

The release validates that required files, columns, and machine-readable summaries exist and remain aligned across the retained branch table, full ledger, GraphML, provenance, validation, and archive-control surfaces. The schema package distinguishes identifiers, observed and derived values, provenance metadata, validation variables, and explicit missing electrical fields.

For the frozen public-validation route, the internal validation export records 25 machine-readable checks across source geometry parsing, retained-branch completeness, ledger accounting, endpoint summaries, facility footprints, GraphML/table parity, sensitivity rows, and artifact hashes. The result is PASS_WITH_WARNINGS: 23 checks pass and two checks are release-review warnings rather than data-consistency failures. The release geometry and processing workspace are now documented in Data Records, while the exact wording of the upstream portal CRS remains unverified. Clean-room extraction validation of the v1.0.1 archive reports no manifest, checksum, dictionary-coverage, or headline-count failures. The earlier v1.0.0 tag rebuild remains the evidence for byte-for-byte reconstruction from frozen derived inputs; a v1.0.1 tagged rebuild is still required, and raw/API reacquisition remains subject to the limitation stated in Usage Notes. The same export records 5,334 valid source geometries and zero invalid/dropped geometries; zero missing required retained-branch fields; zero missing required ledger fields; 358 retained branches; 984 downgraded or rejected records; 1,342 ledger rows; 64,008 endpoint-index rows across six snap thresholds; and 216 sensitivity rows.

5.4 Referential integrity

Referential checks confirm that endpoint facility references, GraphML node and edge identifiers, branch-level validation records, source manifests, and manifest-linked artifact relationships resolve to valid upstream records. Graph and table views are checked for parity so that the GraphML, retained branch table, and selected facility set remain consistent with the documented isolate policy.

5.5 Graph consistency

The selected all-facility graph contains 484 facilities, 358 multiedges, 170 connected components, a largest component of size 53, 109 isolates, and 28 parallel edges. The edge-induced graph contains 375 incident facilities, while the all-facility GraphML additionally preserves selected isolates. This accounting distinction is stored explicitly because graph artifacts with different isolate policies can otherwise appear inconsistent. Graph-level summaries therefore validate the released artifact against its own declared representation rather than against an external ground-truth network.

The GraphML/table parity check confirms that all 358 retained branch identifiers appear as GraphML edge identifiers, that no extra GraphML branch identifiers are present, and that every retained branch endpoint facility resolves to a GraphML node. The GraphML has 27 parallel facility pairs, 55 edge records participating in parallel pairs, and 28 extra parallel-edge records beyond one edge per facility pair. The missingness export also confirms that all retained-branch electrical fields `r`, `x`, `b`, thermal limits, transformer impedance, and tap settings remain explicitly marked as MISSING_NOT_ESTIMATED; these missing values are part of the release semantics rather than silent nulls.

5.6 Sensitivity analysis

Sensitivity analysis addresses whether the selected release point is stable under the declared reconstruction design space. The direct endpoint-to-substation baseline yields 51 merged edges and a six-node largest component, whereas the selected reconstruction yields 358 retained edges and a 53-node largest component for the current snapshot. These values describe yield and connectivity under two workflows; they are not treated as external accuracy improvements.

Across the 216-setting sweep, increasing the facility buffer from 100 m to 500 m at a 0.5 m endpoint snap threshold raises retained branches from 358 to 369 but also increases ambiguity from 61 to 89 and reduces the largest component from 53 to 27. At 50 m endpoint snapping, only three inter-facility circuits remain for each tested buffer. Figure 4 shows these trade-offs and the selected operating point. These results support the claim that reconstruction outputs are sensitive to plausible preprocessing settings rather than identifying a uniquely correct solution.

5.7 External triangulation

The external triangulation workflow searches for branch-level public evidence outside the selected reconstruction rule. The current implementation uses OpenStreetMap/OpenInfraMap 60 kV `power=line` and `power=cable` records in Portugal [OpenStreetMap contributors, 2026, OpenInfraMap contributors, 2026], parses voltage, operator, name, reference, circuit, and geometry fields, and compares each retained branch by endpoint-name overlap and corridor proximity. A name score must be at least 0.8. Strong spatial evidence requires minimum distance no greater than 250 m, median branch-to-OSM distance no greater than 500 m, and branch or OSM coverage of at least 0.5 within 500 m; weak proximity is limited to 1,000 m. The operator condition requires a normalized tag containing `e redes`.

Table 4 gives the category counts. For the current snapshot, 5,551 Portuguese 60 kV public ways were evaluated and all 358 retained candidate branches receive a non-empty public-evidence category, with 247 strong, 104 medium, and seven weak matches. This result provides public-source concordance and a branch-level evidence index. It does not establish operator validation or branch accuracy, because OSM coverage and tagging are non-authoritative. The internal independence audit inspected histories for the 271 unique matched OSM ways, covering 2,326 historical versions, and classifies 266 retained branches as having more-independent public evidence, 87 as unknown independence, and five as possibly same-source. The public archive contains only summary counts, evidence tables, and a sanitized branch-level audit; raw caches, user identifiers, changesets, and element-history dumps are excluded. These categories must not be interpreted as precision, recall, or confirmation of every retained branch.

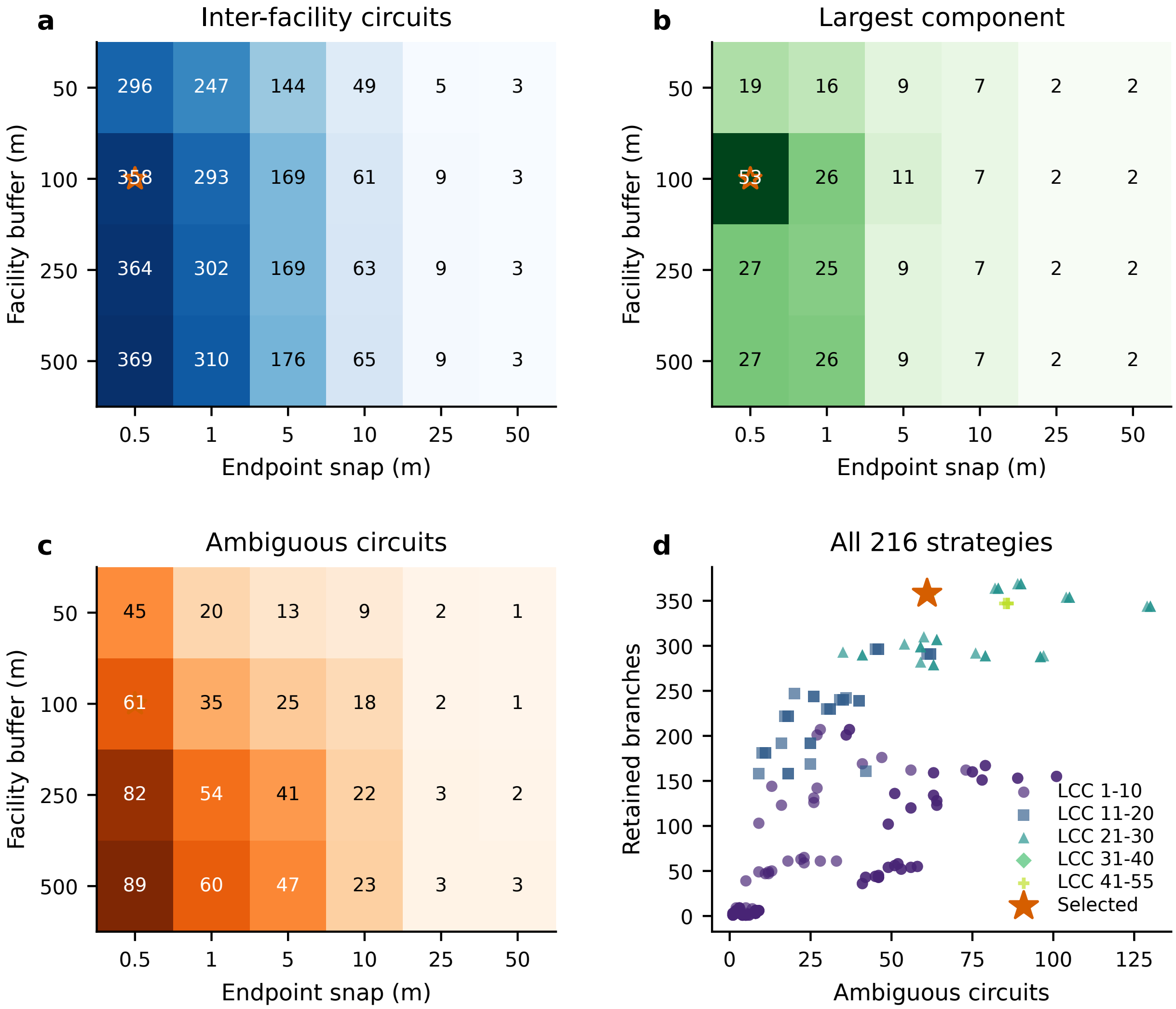

Figure 4: Sensitivity of retained-branch counts, ambiguity, and graph connectivity to facility buffers and endpoint snap thresholds. The marked operating point indicates the documented selected setting for the release; it is not an externally calibrated optimum.

5.8 Claim-bounded validation strategy

The release does not use human-adjudicated truth labels and does not report an accuracy metric. The validation argument is deliberately claim-bounded. Internal tests establish that the released records satisfy their declared schema, classification, provenance, and graph contracts. The 216-setting sweep quantifies sensitivity to reconstruction choices. The external comparison records concordance with a separately maintained public geospatial layer while preserving the evidence category and source caveat for every retained branch. Together, these tests support reproducibility, internal consistency, sensitivity transparency, and public-source triangulation; they do not support a claim that the inferred connections reproduce the operator's true network.

The public-source comparison is evaluated at branch level but is not converted into binary ground truth. Consequently, the manuscript reports category counts and matching variables rather than precision, recall, F_1, or confidence intervals. Precision is not independently estimated, and national or real-network recall is not estimable because neither an authoritative branch census nor a probability sample of true Portuguese 60 kV connections is available. Table 5 states the supported and excluded interpretation for each validation

Table 4: OSM/OpenInfraMap public-source triangulation for the 358 retained candidate branches. Categories describe review-prioritization evidence, not truth labels.

Evidence category	Branches	Interpretation
OSM_NAME_OPERATOR_STRONG	183	Endpoint names appear in a 60 kV E-REDES-labelled OSM record.
OSM_GEOMETRY_OPERATOR_STRONG	64	Candidate corridor is close to a 60 kV E-REDES-labelled OSM corridor.
OSM_PARTIAL_NAME_NEARBY	56	One endpoint name appears and a nearby 60 kV corridor is present.
OSM_GEOMETRY_MEDIUM	48	Candidate overlaps a 60 kV public corridor without strong name evidence.
OSM_NEARBY_WEAK	7	A nearby 60 kV public feature is present, but evidence is incomplete.
NO_EXTERNAL_OSM_MATCH	0	No retained branch falls in this class for the documented snapshot.

Table 5: Claim-bounded validation used in the public-source validation route. "Supported" refers to the stated dataset property, not to physical topology truth.

Validation layer	Supported interpretation	Interpretation explicitly excluded
Geometry, CRS, schema, and key checks	Machine-readable and internally coherent release	Correct physical connectivity
Ledger and graph/table parity	Complete accounting under declared rules	Completeness of the real grid
216-setting sensitivity sweep	Dependence on documented reconstruction choices	Externally optimal parameter setting
OSM/OpenInfraMap triangulation	Public-source name and corridor concordance	Operator validation or statistical accuracy
Electrical-readiness gates	Explicit identification of missing or diagnostic fields	AC power-flow or operational readiness

layer.

Two deterministic negative controls test whether the public-source matcher responds to deliberately broken evidence channels. First, an endpoint-name control was run on all 358 retained branches. The control keeps each branch geometry, voltage, status, length, confidence score, and branch identifier fixed, but replaces the endpoint-name pair with a length-stratified cyclic shift from another retained branch using seed 20260714. Under the unmodified endpoint names, 183 of 358 branches receive strong name-based OSM/OpenInfraMap evidence. Under corrupted endpoint names, this falls to 9 of 358 branches, a 95.1% relative reduction. Second, a spatial-alignment control keeps endpoint names and tabular branch attributes fixed but translates every branch geometry by +1.75 degrees longitude and +0.85 degrees latitude. All 358 transformed geometries remain valid and no records are excluded after transformation. Under the unmodified geometries, 290 of 358 branches satisfy the corridor-evidence criterion; under displaced geometries, this falls to 1 of 358 branches, a 99.7% relative reduction. These controls support the selectivity of the endpoint-name and corridor components of the matcher. They do not estimate topology accuracy, precision, recall, or operator validation.

5.9 Validation boundaries

The current validation evidence supports internal consistency, release transparency, sensitivity analysis, and public-source triangulation. It does not support claims of operator-confirmed topology accuracy, independently estimated branch precision, national-grid recall or completeness, or electrical-operating realism. Optional electrical-readiness artifacts remain diagnostic because branch-specific parameters, transformer details, control states, measured injections, and verified boundaries are incomplete or scenario-derived. The full diagnostic AC power-flow extension therefore remains outside the validated core dataset and should not be used to imply AC power-flow or OPF readiness.

6 Usage Notes

6.1 Intended uses

The core release is intended for topology-reconstruction studies, geospatial data integration, provenance-aware network analysis, uncertainty analysis, dataset-engineering experiments, and development of downstream graph or tabular interfaces that preserve release semantics. The retained branch table and circuit-classification ledger should be used together whenever a study depends on understanding what was retained, what was downgraded, and why.

6.2 Non-intended uses

The release is not intended for operational switching studies, protection analysis, security assessment, contingency analysis, congestion forecasting, infrastructure-targeting applications, emergency operations, asset-condition assessment, commercial network-capacity claims, regulatory network-capacity claims, or claims about the actual Portuguese grid. The candidate topology should not be described as operator-validated, and the optional diagnostic electrical and learning layers should not be described as measured system behavior.

6.3 Reading provenance and semantic status fields

Users should read semantic status fields before aggregating or modeling the records. Observed, derived, inferred, source-backed, scenario-assumed, benchmark-only, proxy, and missing fields represent distinct evidence types. Studies that require source-backed physical parameters or operator-confirmed connectivity should filter the release accordingly and should not treat inferred or scenario-derived fields as interchangeable with observations.

6.4 How to use GraphML

The GraphML is the most direct representation of the released candidate multigraph for graph analysis. It preserves the selected all-facility node set, including isolates, and therefore aligns with the documented component and degree summaries. Users comparing the GraphML with edge-induced views should account for the explicit isolate policy. Parallel edges are part of the released graph structure and should not be collapsed without documenting the transformation.

6.5 How to use CSV records

The retained candidate branch CSV is the preferred branch-level topology table. The circuit-classification ledger should be consulted whenever a downstream workflow depends on failure modes, ambiguity, or alternative selection rules. Validation CSV and JSON records document evidence categories, matcher controls, independence-risk categories, missingness, and archive-validation relationships. The six recommended joins in `schema/join_relationships.csv` should be used in preference to name-based joins.

6.6 Limitations

The release has four main limitations. First, it is a candidate topology derived from public E-REDES records rather than an operator-confirmed network model. Second, external triangulation provides public-source concordance categories rather than truth labels: precision is not independently estimated, and national or real-network recall cannot be estimated from the available records. Third, electrical parameters, transformer details, operating controls, and injections are incomplete or absent, so the release is not AC power-flow or OPF ready. Fourth, the DOI archive preserves the versioned derived dataset but not the raw E-REDES snapshots. Source identifiers, URLs, access timestamps, roles, and attribution are recorded, but the three topology-critical v2.1 export URLs returned HTTP 404 during the final clean-room reacquisition audit; raw/API-to-release byte reproduction is therefore not claimed.

Data Availability

PT60-Candidate version 1.0.1 is prepared for the reserved figshare DOI `https://doi.org/10.6084/m9.figshare.32984021`. The file `PT60-Candidate-v1.0.1.tar.gz`, SHA-256 `a4dd2d115caae0cd1c873fb7db3341e0aaed0ce62f7f0072ae92642898efb69a`, must replace the earlier uploaded draft before publication. It contains the core topology, full circuit ledger, GraphML, sensitivity sweep, provenance, validation, corrected schema/data dictionary, licensing, attribution, manifest, and checksum records described in Data Records. Raw source downloads and optional diagnostic/interface artifacts are excluded as documented in `excluded\_artifacts.json`. E-REDES-derived records remain subject to the attribution and indication-of-modification terms documented in `DATA\_LICENSE.md` and `ATTRIBUTION.md`; the software licence does not replace those data terms.

Code Availability

The source code is publicly available at `https://github.com/MingLeiZhou/PortugueseOPD`. The existing v1.0.0 tags preserve the prior release; an immutable v1.0.1 source tag and DOI-providing software archive remain to be created before submission. The reference environment uses Python 3.12.9 and the versioned `requirements.txt`; the software is MIT licensed. The dataset archive is independently usable without installing the generation pipeline, but full regeneration from raw E-REDES/API responses is subject to the source-reacquisition limitation described in Usage Notes.

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